

Performance Analysis of the Conditional Autoencoder Factor

Empirical Evaluation with Alphalens

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Outline

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What Are We Evaluating?

The Core Question

Does the conditional autoencoder (CAE) produce a **meaningful cross-sectional signal** for predicting stock returns?

The model in one sentence:

- The CAE learns **nonlinear latent factors** from 14 firm characteristics (momentum, liquidity, risk) and produces a **predicted return** for each stock every week.

Evaluation framework:

- Out-of-sample period: **2015 – 2019** (5 rolling folds).
- Forward-return horizons: **5, 10, and 21 trading days**.
- Universe: $\sim 3\,430$ US equities at weekly frequency.
- Tool: **Alphalens** (quantile sorting, IC, turnover diagnostics).

How the Test Works

Sorting procedure (every week):

- 1 Rank all stocks by the CAE predicted return.
- 2 Split into 5 equal-weight quantile portfolios:
Q1 = lowest 20 % → **Q5** = highest 20 %.
- 3 Measure *realised* forward returns for each group.

Key test:

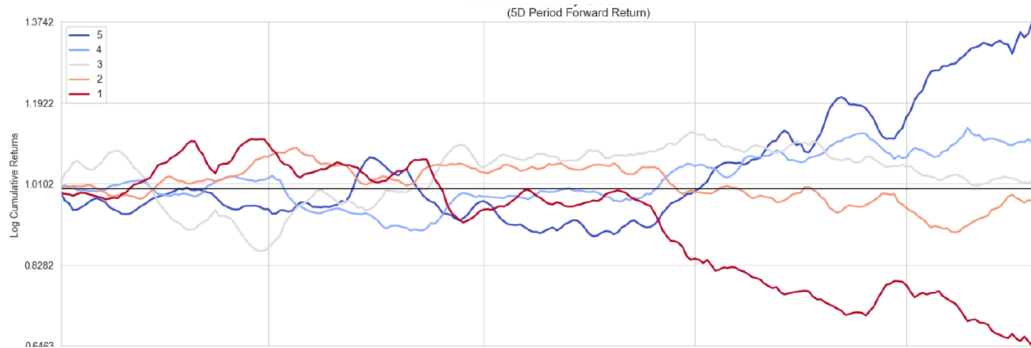
If the factor is informative, Q5 should *outperform* Q1 consistently $\Rightarrow$ monotonic ordering.

Table: *

Quantile Statistics	Q	Mean	Std
	1	-0.097	0.146
	2	-0.042	0.097
	3	-0.019	0.073
	4	+0.004	0.054
	5	+0.046	0.044

- ✓ **Monotonic** mean from Q1 to Q5.
- ✓ **Decreasing** std $\Rightarrow$ top quantile is more stable.

Cumulative Returns by Quantile (5-Day Horizon)



Reading the plot: Q5 (dark blue) rises steadily; Q1 (dark red) declines.

⇒ The CAE correctly ranks stocks: *higher predicted return = higher realised return.*

Mean Returns by Quantile (All Horizons)

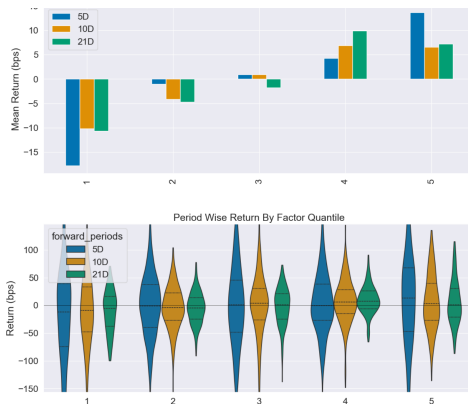


Figure 3.6: Mean and period-wise returns by factor quantile for 5-, 10-, and 21-day horizons. The top panel reports mean returns; the bottom panel displays the full return distribution using violin plots.

Top (bars): Monotonic pattern Q1–Q5 at every horizon. **Bottom (violins):** Dispersion narrows for higher quantiles $\Rightarrow$ more stable signal at the top.

Alpha and Beta of the Factor

Table: *

Return Analysis			
	5D	10D	21D
Ann. Alpha	3.05%	0.70%	3.44%
Beta	0.062	0.054	0.009
Top Q (bps)	13.6	6.6	7.2
Bot. Q (bps)	-17.8	-10.2	-10.7
Spread	31.4	16.8	17.8

What This Means

- **Positive alpha** ($\approx 3\%$): factor earns returns *not explained* by the market.
- **Beta ≈ 0** : the signal is *orthogonal* to the market.
- **Long-short spread** of ~ 31 bps/week at 5D, strongest at short horizons.

Long-Short Spread Over Time

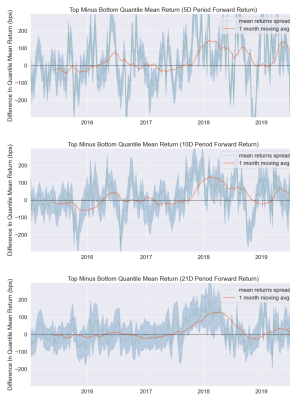


Figure 3.7: Time-series evolution of top-minus-bottom quantile mean returns across 5-, 10-,

Shaded area = weekly spread variability; orange line = 1-month moving average.
⇒ Broadly **positive throughout**, with some attenuation during late-2018 volatility.

Information Coefficient (IC) — Overview

Definition

The IC is the **Pearson correlation** between factor scores at time t and realised returns at $t+h$. Positive IC = higher scores $\Rightarrow$ higher future returns.

	5D	10D	21D
IC Mean	0.008	0.004	0.009
IC Std.	0.053	0.047	0.041
Risk-adj. IC	0.146	0.095	0.229
t -stat	2.27	1.47	3.54
p -value	0.024	0.142	0.000

Interpretation

- IC is **positive at all horizons**: signal points in the right direction.
- **Statistically significant** at 5D ($p=0.024$) and 21D ($p<0.001$).
- Magnitudes ~ 0.004 – 0.009 are **typical** for equity factors — small but valuable.
- **Decreasing std** at longer horizons: more stable signal.

IC Distribution and Normality Check

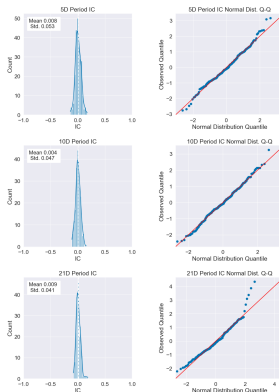


Figure 3.8: Distribution and Q-Q plots of the Information Coefficient across horizons. Each

Left: Symmetric histograms centred near zero with a slight positive shift. **Right:** Q-Q plots align with the normal line $\Rightarrow$ IC is **well-behaved**.

Rolling IC Over Time



Figure 9.6: Rolling Information Coefficient time series for 5D, 10D, and 21D forward returns

Blue = weekly IC; green = 1-month moving average. $\Rightarrow$ Recurrent positive phases; no structural breakdowns across market regimes.

Monthly IC Heatmaps

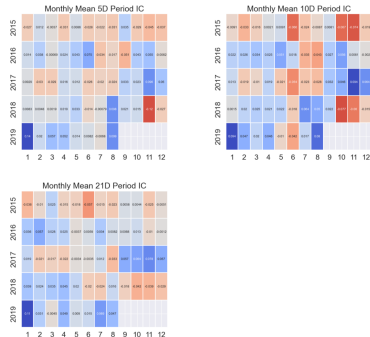


Figure 3.10: Monthly mean Information Coefficients (ICs) for 5D, 10D, and 21D horizons from 2015-2019. Blue tones indicate positive predictive strength, while red tones correspond to temporary signal inversion.

Blue = positive IC; Red = temporary signal inversion. $\Rightarrow$ Persistent **blue regions** at 21D; the factor retains predictive strength across years.

Turnover and Rank Autocorrelation

Table: *

Turnover & Persistence			
	5D	10D	21D
Q1 Turnover	0.50	0.57	0.64
Q5 Turnover	0.49	0.57	0.64
Rank Autocorr.	0.51	0.40	0.29

Turnover $\approx$ 50–65 %: a meaningful fraction of stocks stays in each quantile week-to-week.

Rank autocorrelation decays from 0.51 to 0.29, reflecting natural signal dissipation.

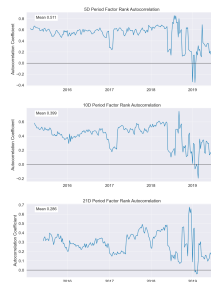


Figure 3.5: Factor rank autocorrelation of the autoencoder-implied factor across 5-, 10-, and 21-day horizons.

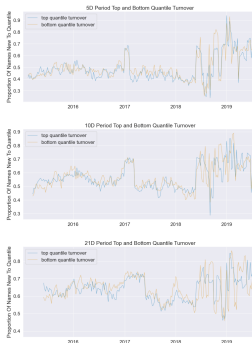
From a financial perspective, this joint analysis highlights an appealing balance between adaptability and persistence. The conditional autoencoder updates frequently enough to capture evolving market information, yet its factor structure remains sufficiently stable to

Factor rank autocorrelation over time.
 $\Rightarrow$ Positive and stable: the factor **does not reshuffle randomly**.

Quantile Turnover Over Time

quantiles to facilitate comparison, with the argument `axis=1` ensuring that each quantile's series is aligned side by side along the column dimension.

The complete turnover and persistence analysis is then automatically produced using the `create_turnover_tear_sheet` function `al.tears.create_turnover_tear_sheet(factor_data)`, with as result the following two figures and table.



Top and bottom quantile turnover across 5D, 10D, and 21D horizons.
⇒ Stable over time: spikes align with market volatility, not model instability.

Key Takeaways

The conditional autoencoder delivers a valid cross-sectional pricing signal

- ➊ **Monotonic quantile ordering** — Q5 consistently outperforms Q1, with a long–short spread of ~ 31 bps/week at 5D.
- ➋ **Market-orthogonal alpha** — Annualised alpha $\approx 3\%$ with near-zero beta: the factor captures firm-specific content, not market risk.
- ➌ **Statistically significant IC** — Positive at all horizons, significant at 5D ($p=0.024$) and 21D ($p<0.001$), with near-Gaussian distributions.
- ➍ **Temporal stability** — Rolling ICs and monthly heatmaps show persistent positive signal with no structural breakdowns across market regimes.
- ➎ **Reasonable turnover** — 50–65 % weekly turnover with smooth rank-autocorrelation decay: good balance between reactivity and persistence.

The CAE compresses high-dimensional firm information into a nonlinear latent score that orders the cross-section coherently and delivers meaningful predictions.

Thank You

Questions?